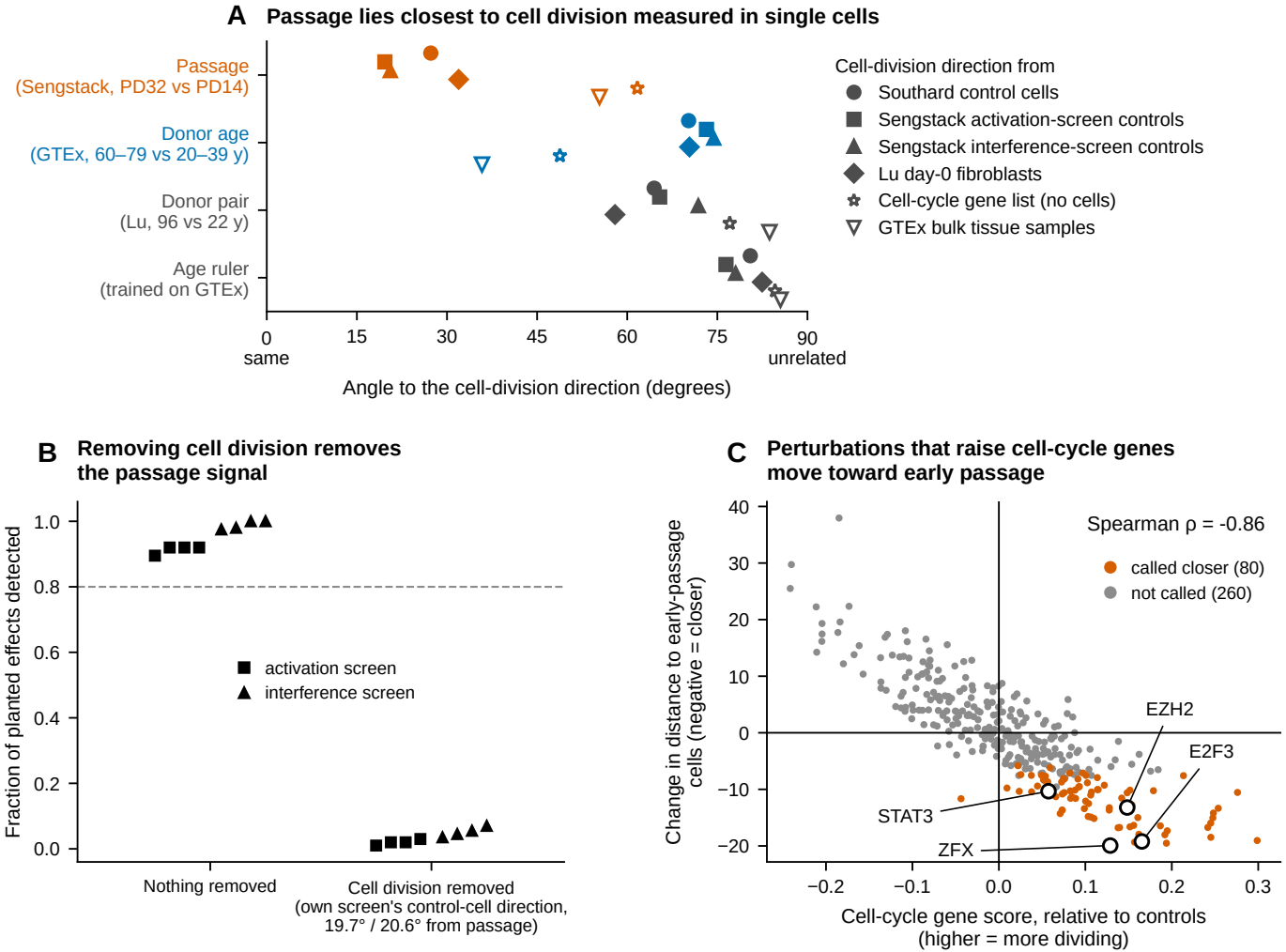
A
Passage lies closest to cell division measured in single cells
Passage
(Sengstack, PD32 vs PD14)
Donor age
(GTEx, 60–79 vs 20–39 y)
Donor pair
(Lu, 96 vs 22 y)
Age ruler
(trained on GTEx)
0
same
15
30
45
60
75
90
unrelated
Angle to the cell-division direction (degrees)
Cell-division direction from
Southard control cells
Sengstack activation-screen controls
Sengstack interference-screen controls
Lu day-0 fibroblasts
Cell-cycle gene list (no cells)
GTEx bulk tissue samples
B
Removing cell division removes
the passage signal
Fraction of planted effects detected
1.0
0.8
0.6
0.4
0.2
0.0
activation screen
interference screen
Nothing removed
Cell division removed
(own screen's control-cell direction,
19.7° / 20.6° from passage)
C
Perturbations that raise cell-cycle genes
move toward early passage
Change in distance to early-passage
cells (negative = closer)
40
30
20
10
0
−10
−20
Spearman ρ = -0.86
called closer (80)
not called (260)
EZH2
E2F3
STAT3
ZFX
−0.2
−0.1
0.0
0.1
0.2
0.3
Cell-cycle gene score, relative to controls
(higher = more dividing)